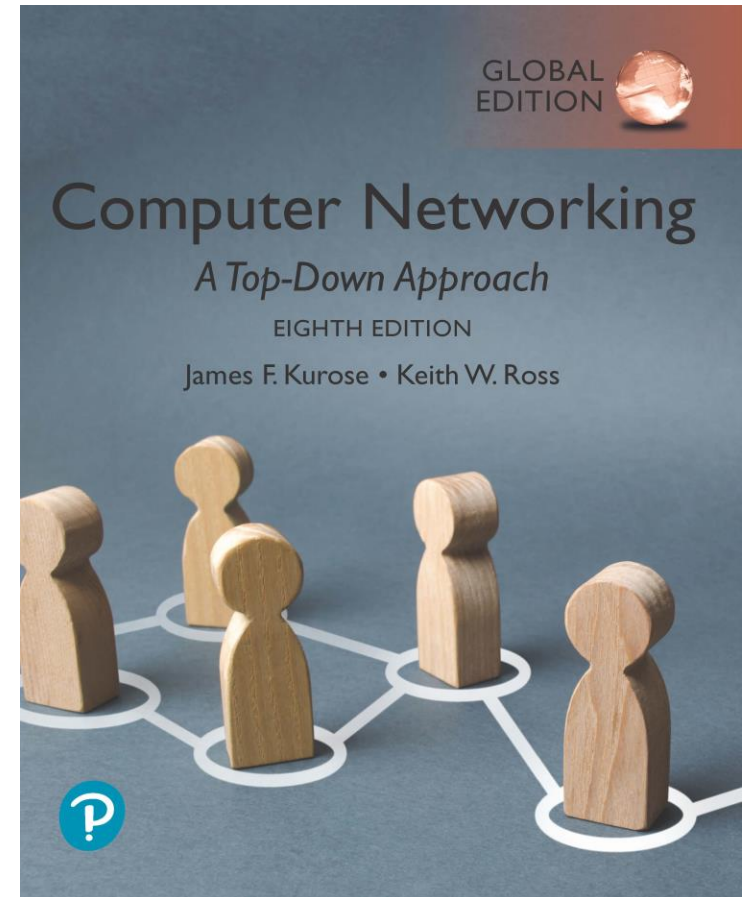


Chapter 7

Wireless and Mobile Networks



Computer Networking: A Top-Down Approach

8th Edition, Global Edition

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Chapter 7 outline

- Introduction

Wireless

- Wireless links and network characteristics
- WiFi: 802.11 wireless LANs
- Cellular networks: 4G and 5G



4G/5G cellular networks

- *the* solution for wide-area mobile Internet
- widespread deployment/use:
 - more mobile-broadband-connected devices than fixed-broadband-connected devices (5-1 in 2019)!
 - 4G availability: 97% of time in Korea (90% in US)
- transmission rates up to 100's Mbps (5G even higher...1Gbps)
- technical standards: 3rd Generation Partnership Project (3GPP)
 - www.3gpp.org
 - 4G: Long-Term Evolution (LTE) standard

4G/5G cellular networks

similarities to wired Internet

- edge/core distinction, both belong to same carrier
- global cellular network: a network of networks
- widespread use of protocols we've studied: HTTP, DNS, TCP, UDP, IP, NAT, separation of data/control planes, SDN, Ethernet, tunneling
- interconnected to wired Internet

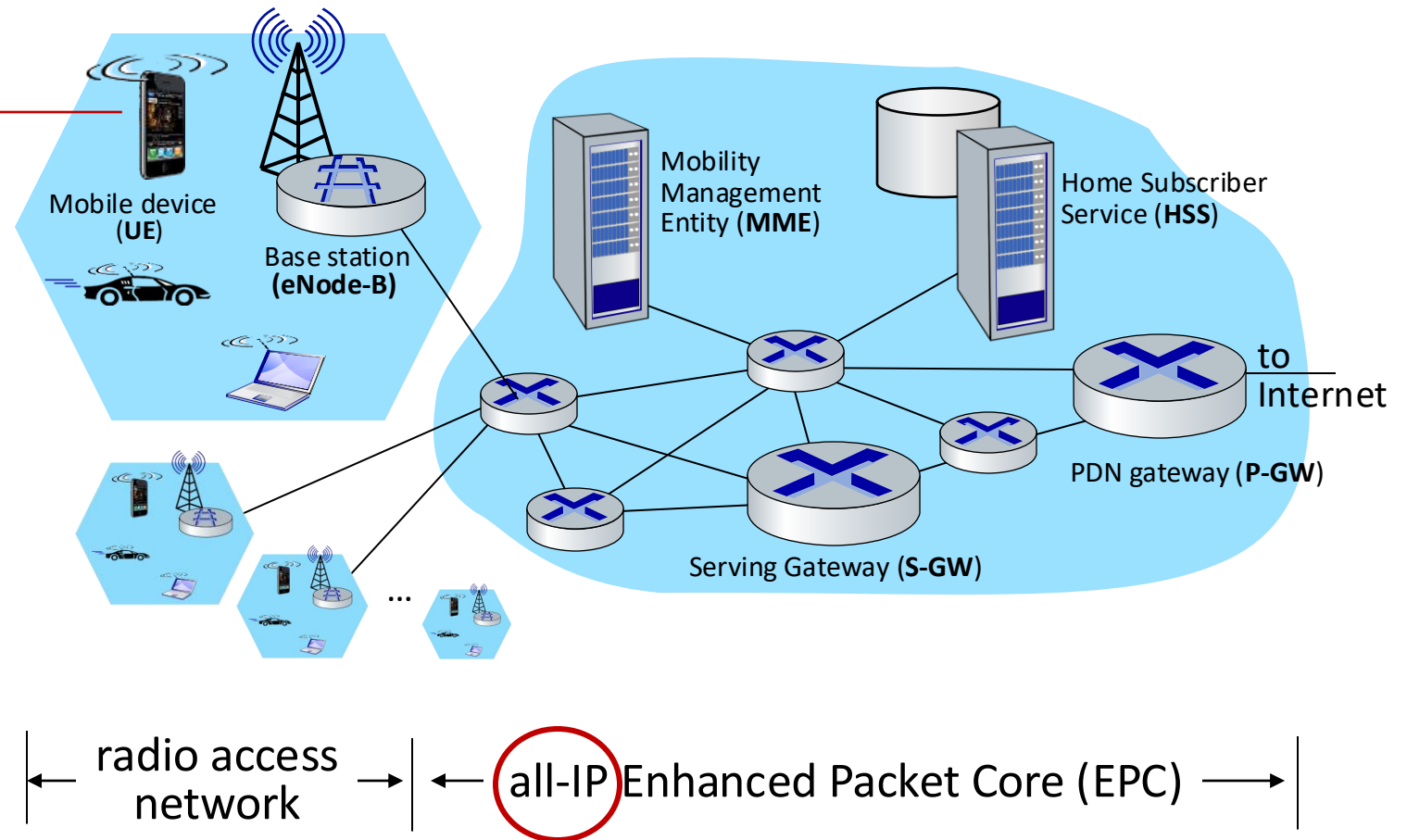
differences from wired Internet

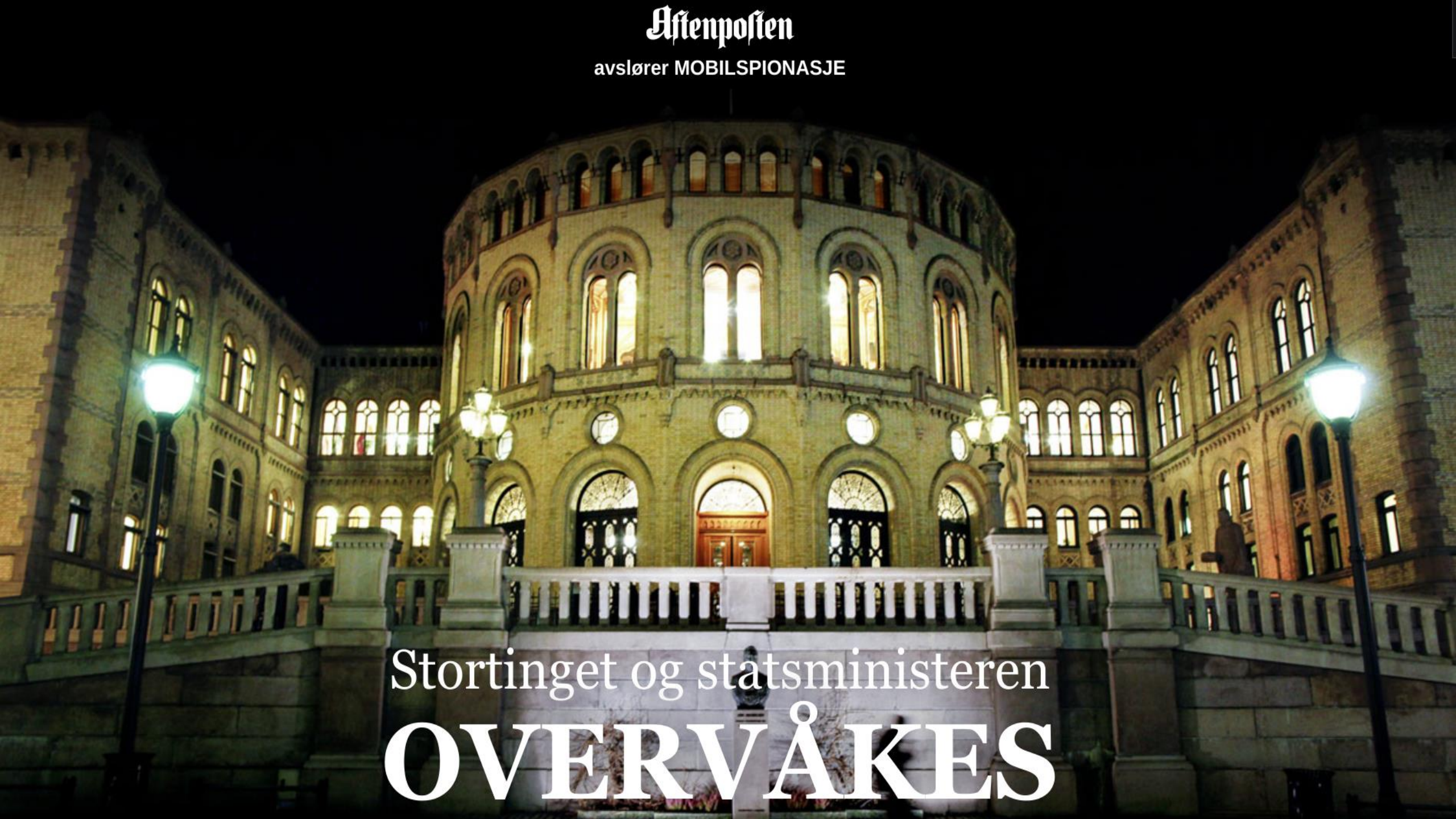
- different wireless link layer
- mobility as a 1st class service
- user “identity” (via SIM card)
- business model: users subscribe to a cellular provider
 - strong notion of “home network” versus **roaming** on visited nets
 - global access, with authentication infrastructure, and **inter-carrier settlements**

Elements of 4G LTE architecture

Mobile device:

- smartphone, tablet, laptop, IoT, ... with 4G LTE radio
- 64-bit International Mobile Subscriber Identity (**IMSI**), stored on SIM (Subscriber Identity Module) card
- LTE jargon: User Equipment (UE)





Aftenposten

avslører MOBILSPIONASJE

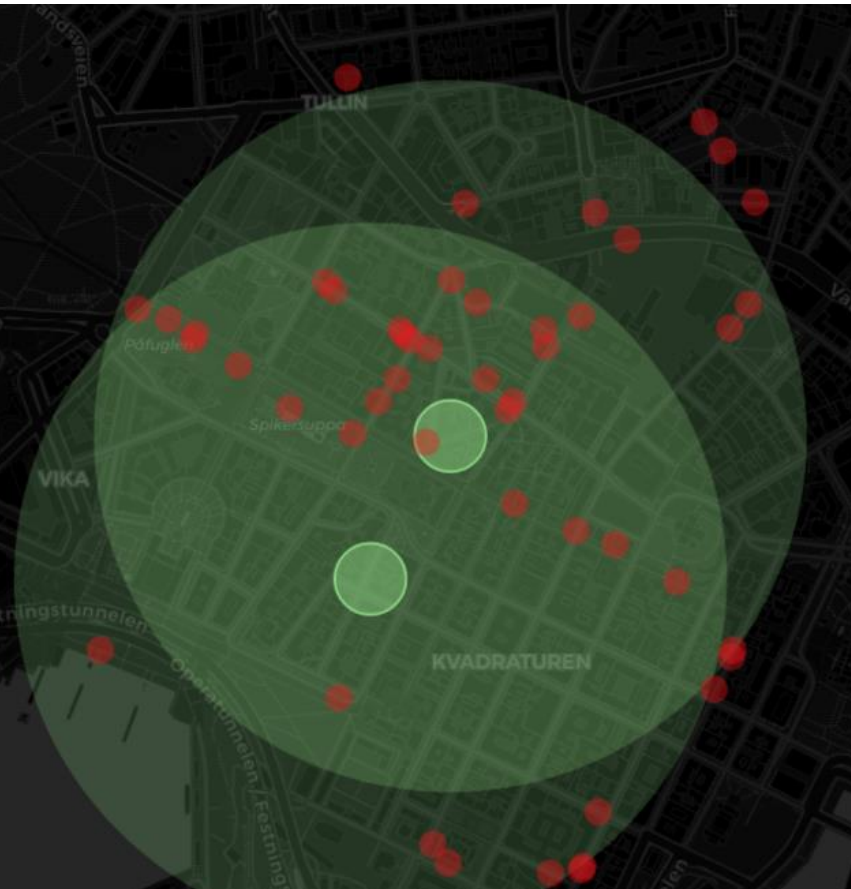
Stortinget og statsministeren
OVERVÅKES

FALSKE BASESTASJONER VED STORTINGET

De **grønne punktene** viser området hvor de falske basestasjonene er plassert. Ca. 50 meter feilmargin.

Dekningsområdet til basestasjonene (ca. 500 meters radius) er de store **grønne sirklene**.

De **røde punktene** viser mistenkelig datatrafikk Aftenposten har registrert i området. Klikk på disse for å lese mer.



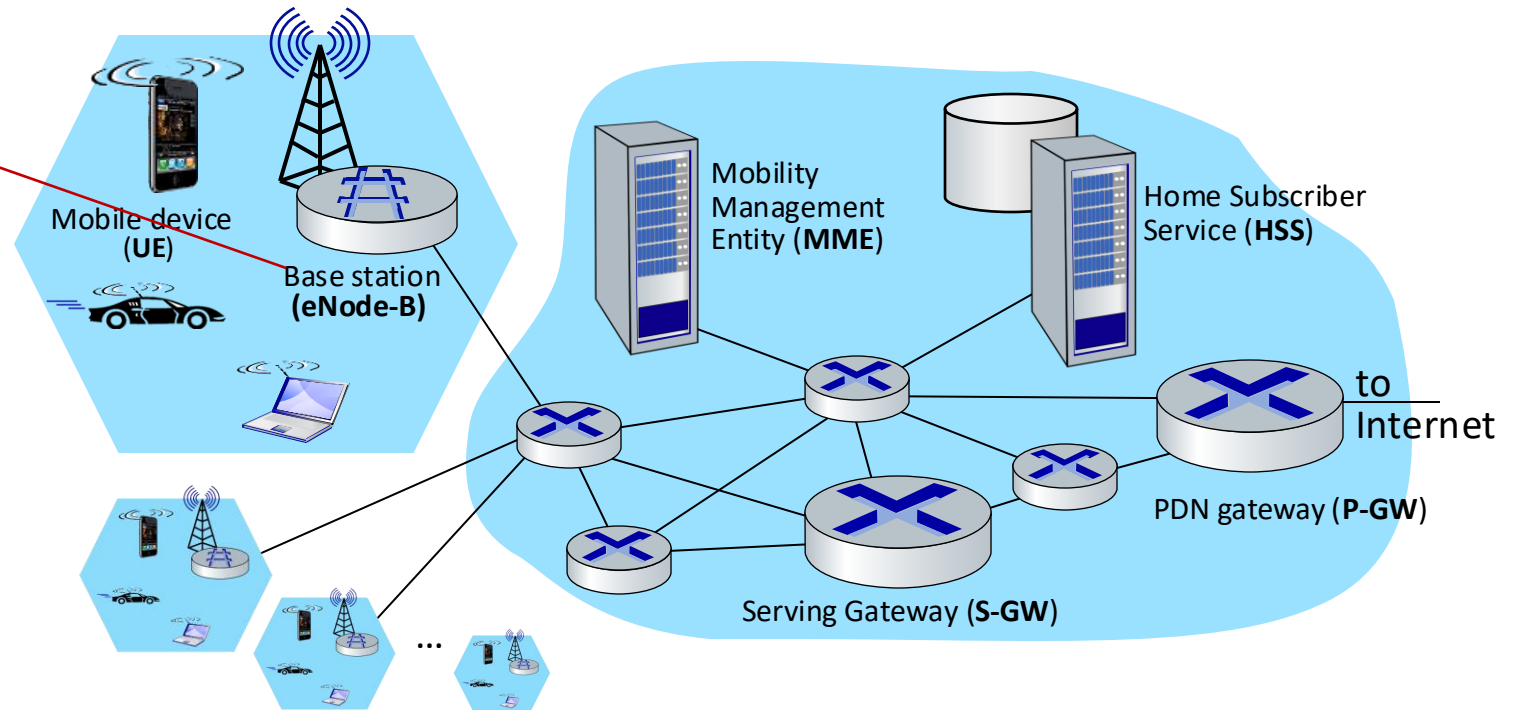
Aftenpostens kartlegging

For første gang i Norge har Aftenposten kartlagt bruken av mobilovervåkingsutstyr i Oslo. Målingene avslører at hemmelige, falske basestasjoner, såkalte IMSI-catchere, høyst sannsynlig er i bruk rundt flere sentrale bygninger i hovedstaden:

Elements of 4G LTE architecture

Base station:

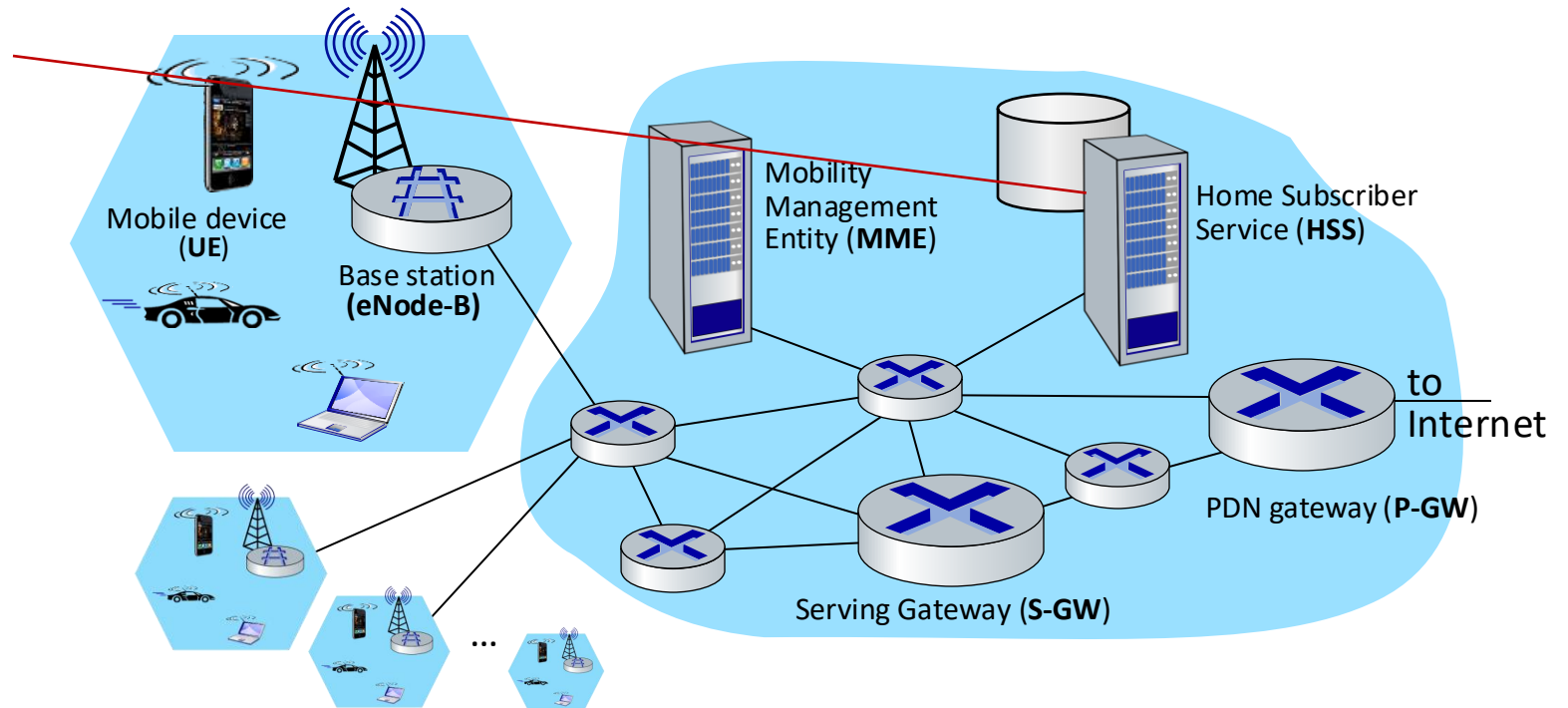
- at “edge” of carrier’s network
- manages wireless radio resources, mobile devices in its coverage area (“cell”)
- coordinates device authentication with other elements
- similar to WiFi AP but:
 - active role in user mobility
 - coordinates with nearby base stations to optimize radio use
- LTE jargon: eNode-B



Elements of 4G LTE architecture

Home Subscriber Service

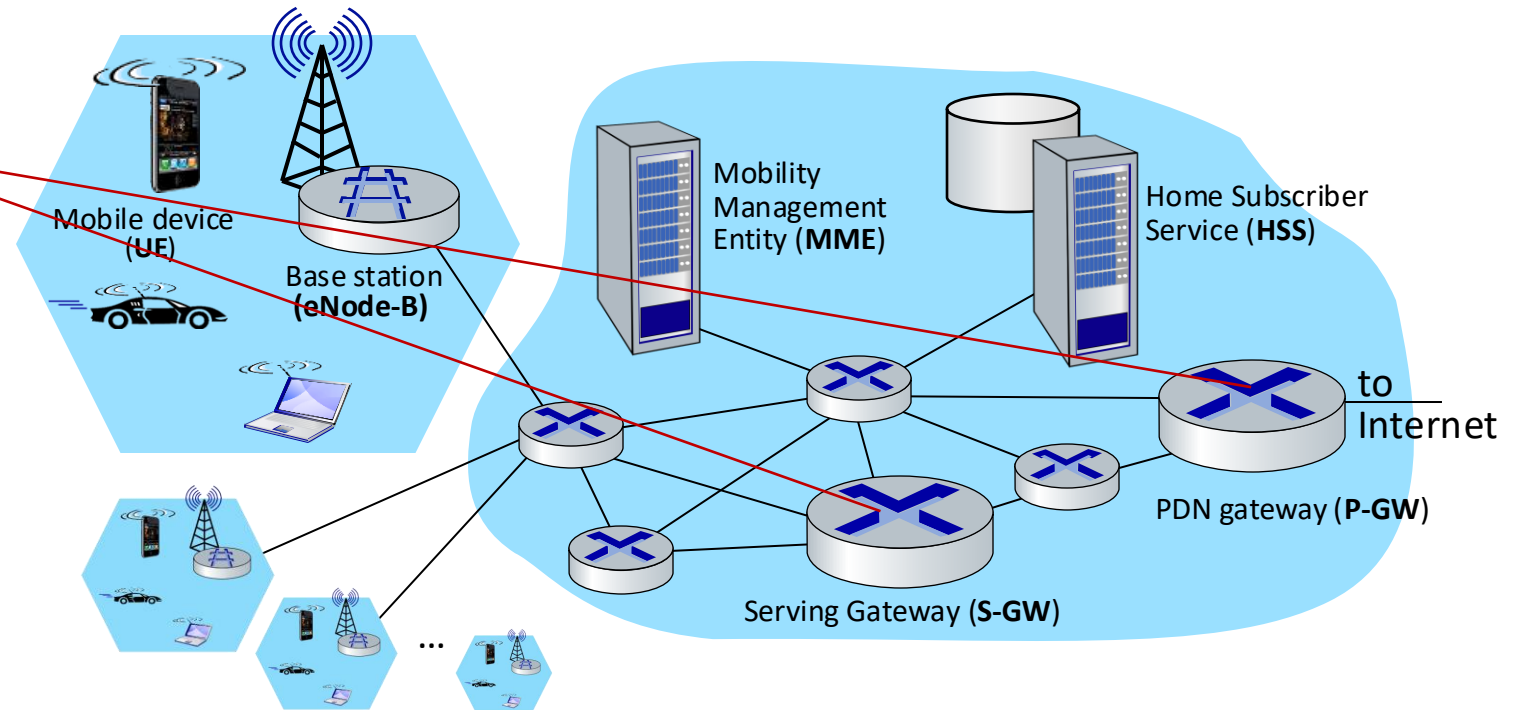
- stores info about mobile devices for which the HSS's network is their “home network”
- works with MME in device authentication



Elements of 4G LTE architecture

Serving Gateway (S-GW), PDN Gateway (P-GW)

- lie on data path from mobile to/from Internet
- P-GW
 - gateway to mobile cellular network
 - Looks like any other internet gateway router
 - provides NAT services
- other routers:
 - extensive use of tunneling

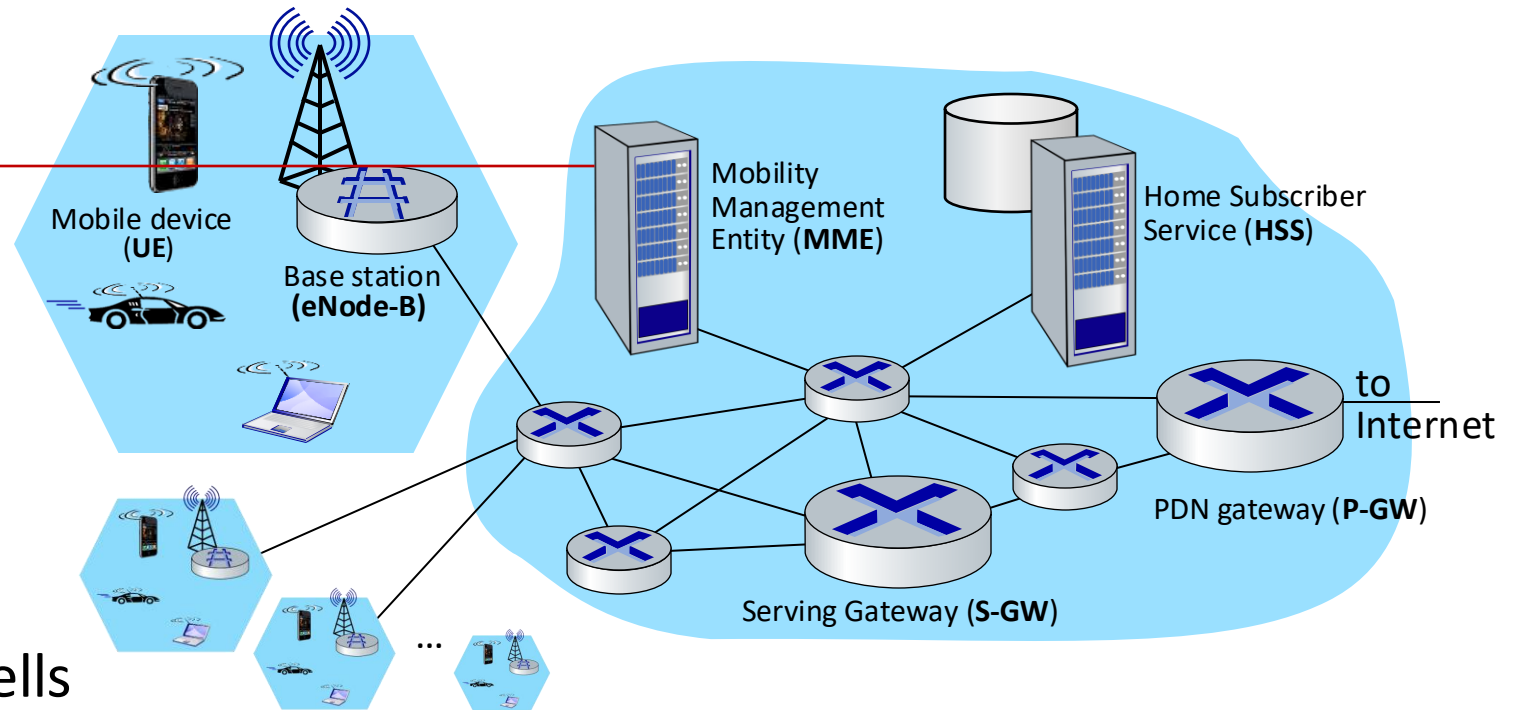


- S-GW—In the LTE architecture, the node that handles all signaling messages to and from the user equipment is called the Serving Gateway (S-GW). (The SGSN in 3G networks is different from the S-GW in 4G networks.).
- P-GW—In 2G and 3G networks, the node that handled all user packets to and from the user equipment is called the GGSN. In the LTE architecture, this is the Packet Gateway (P-GW) or sometimes seen as the Packet Data Network Gateway (PDN-GW).

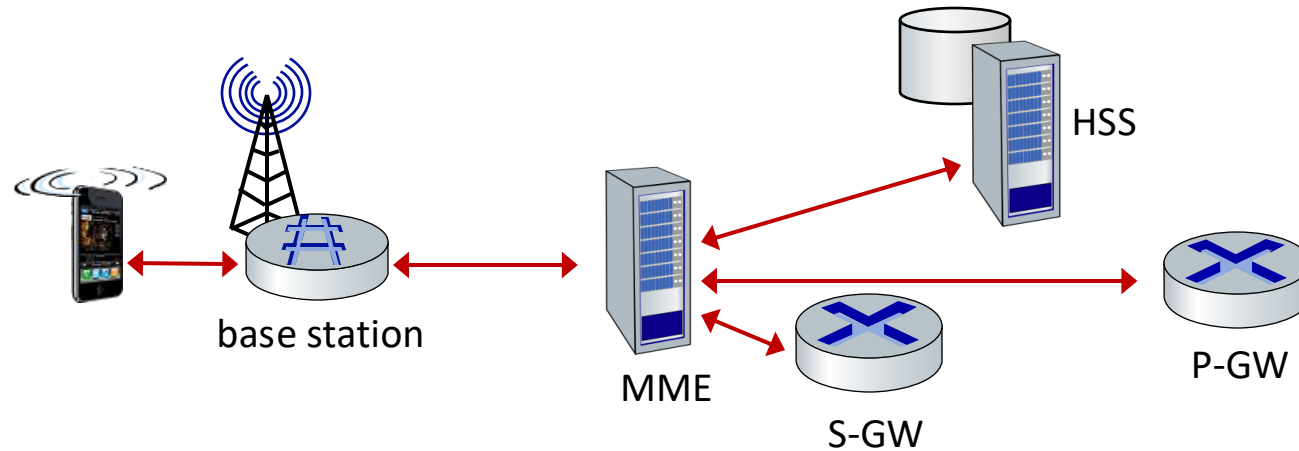
Elements of 4G LTE architecture

Mobility Management Entity

- device authentication (device-to-network, network-to-device) coordinated with mobile home network HSS
- mobile device management:
 - device handover between cells
 - tracking/paging device location
- path (tunneling) setup from mobile device to P-GW

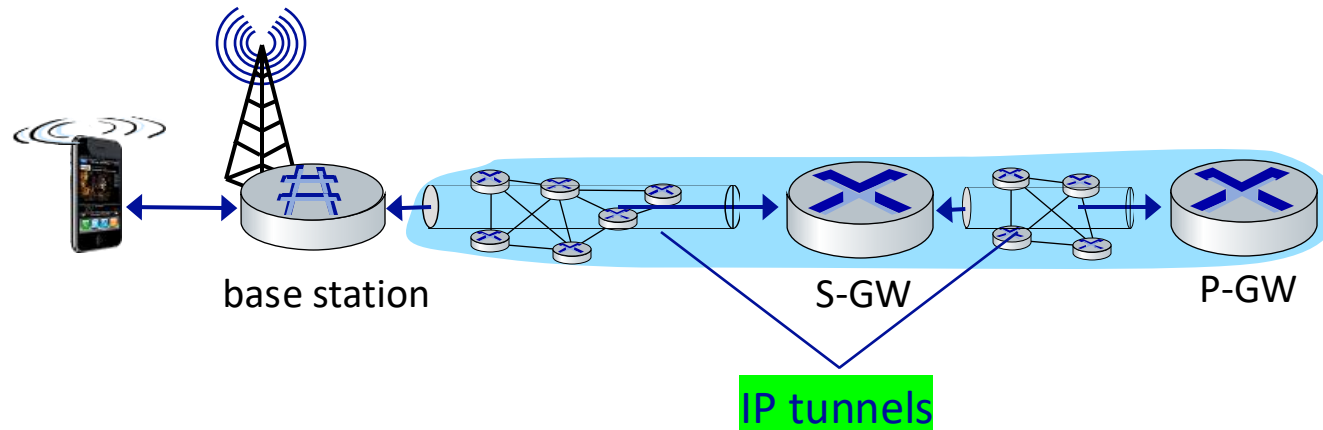


LTE: data plane control plane separation



control plane

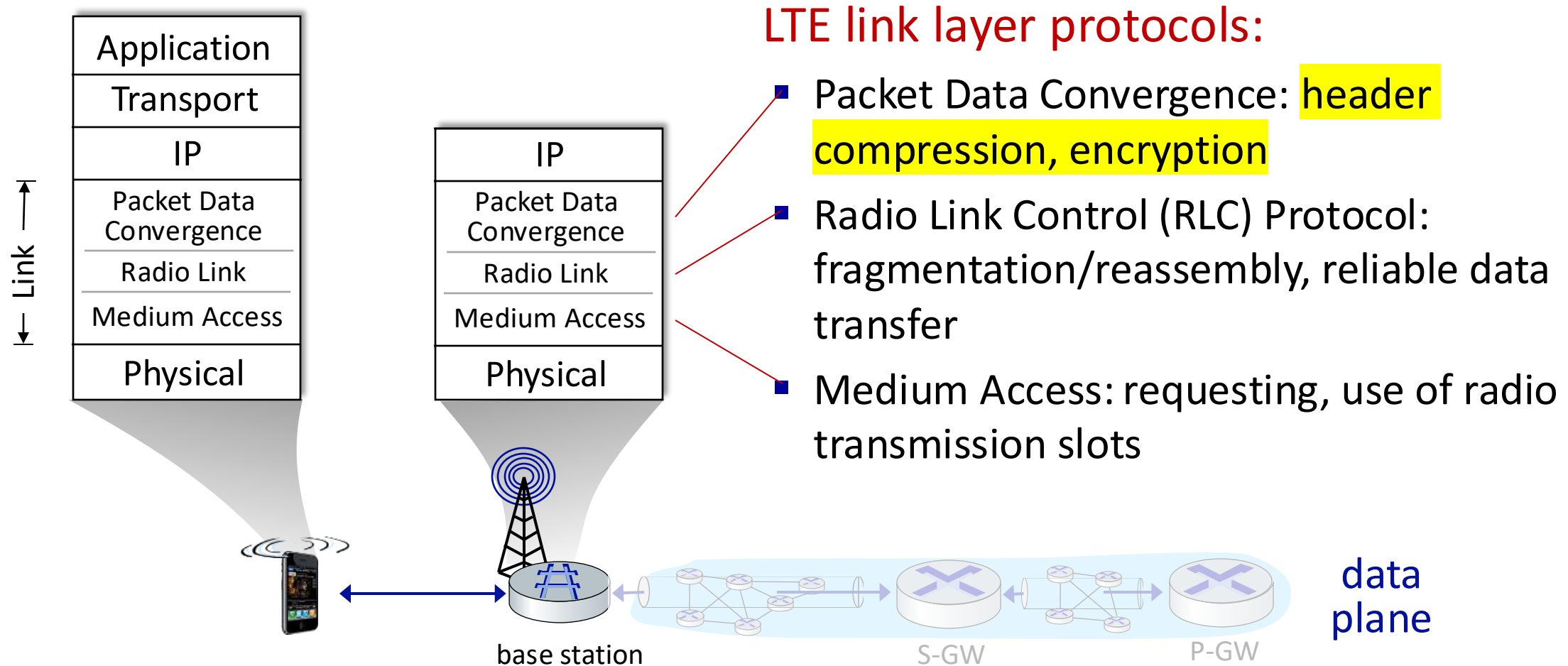
- new protocols for mobility management , security, authentication (later)



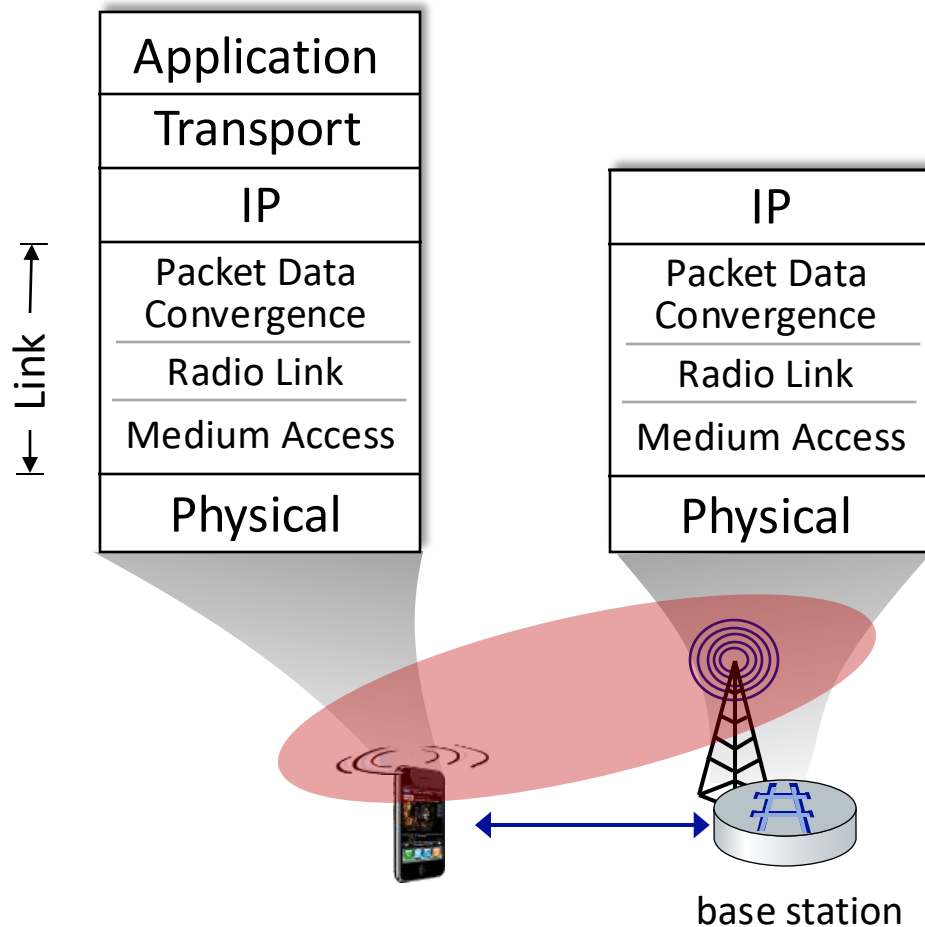
data plane

- new protocols at link, physical layers
- extensive use of tunneling to facilitate mobility

LTE data plane protocol stack: first hop



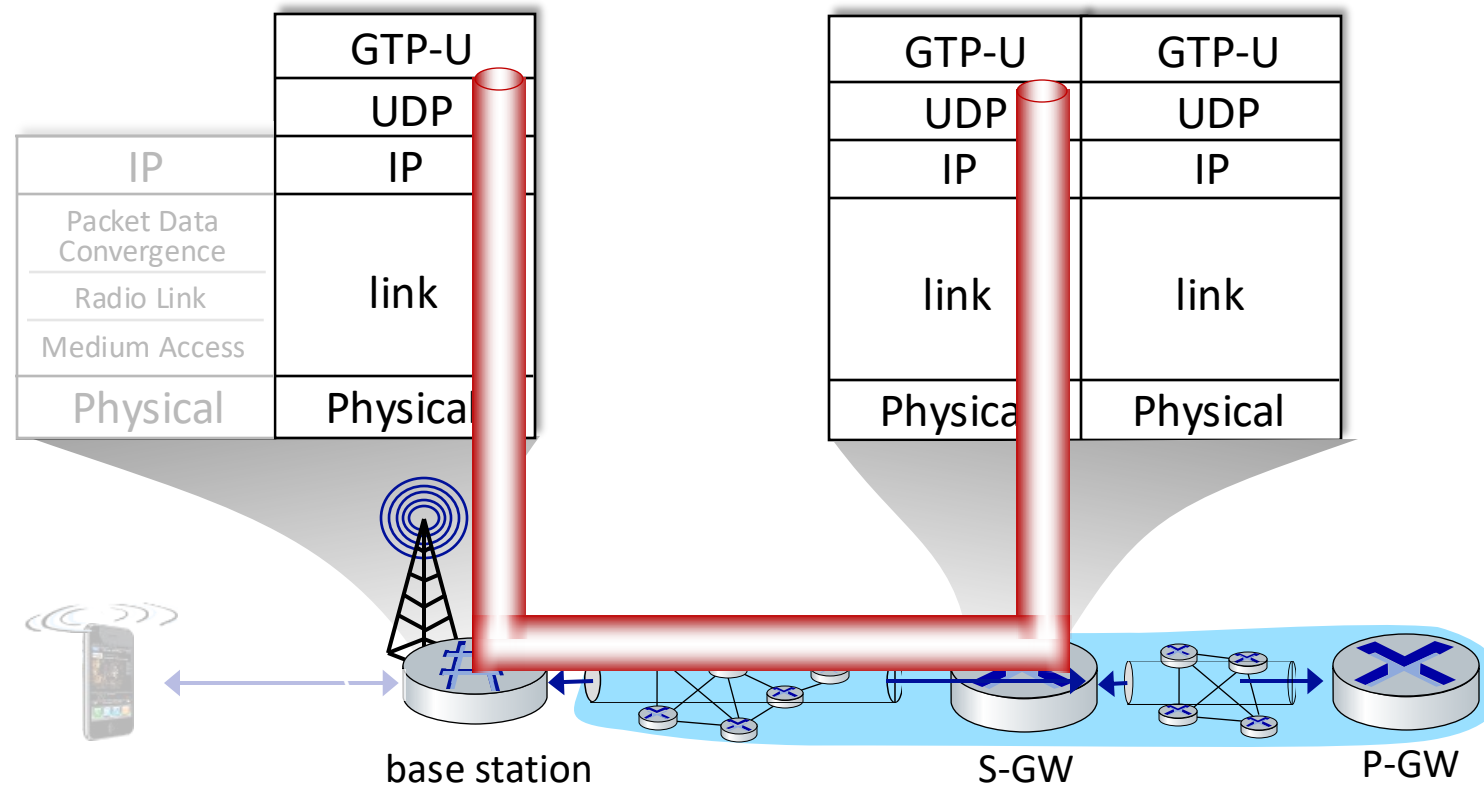
LTE data plane protocol stack: first hop



LTE radio access network:

- **downstream channel:** FDM, TDM within frequency channel (OFDM - orthogonal frequency division multiplexing)
 - “orthogonal”: minimal interference between channels
 - **upstream:** FDM, TDM similar to OFDM
- **each active mobile device allocated two or more 0.5 ms time slots over 12 frequencies**
 - scheduling algorithm not standardized – up to operator
 - 100's Mbps per device possible

LTE data plane protocol stack: packet core

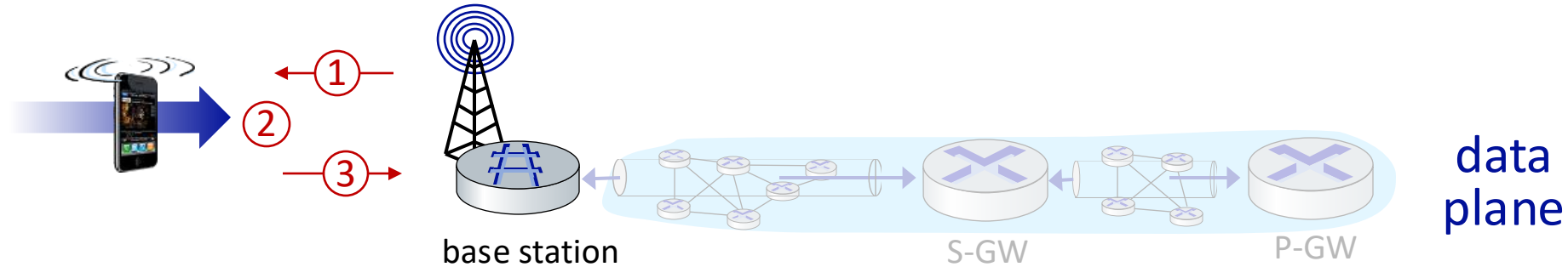


tunneling:

- mobile datagram encapsulated using GPRS Tunneling Protocol (GTP), sent inside UDP datagram to S-GW
- S-GW re-tunnels datagrams to P-GW
- supporting mobility: only tunneling endpoints change when mobile user moves

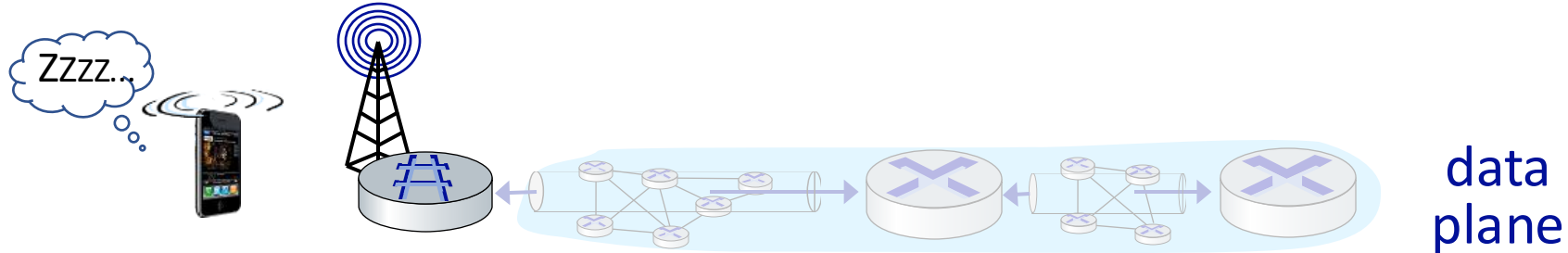
<https://whatismyipaddress.com/>

LTE data plane: associating with a BS



- ① BS broadcasts primary **synch signal** every 5 ms on all frequencies
 - BSs from multiple carriers may be broadcasting synch signals
- ② mobile finds a primary synch signal, then locates **2nd synch signal** on this freq.
 - mobile then finds info broadcast by BS: channel bandwidth, configurations; BS's cellular carrier info
 - mobile may get info from multiple base stations, multiple cellular networks
- ③ mobile selects which BS to associate with (*e.g.*, preference for home carrier)
- ④ more steps still needed to authenticate, establish state, set up data plane

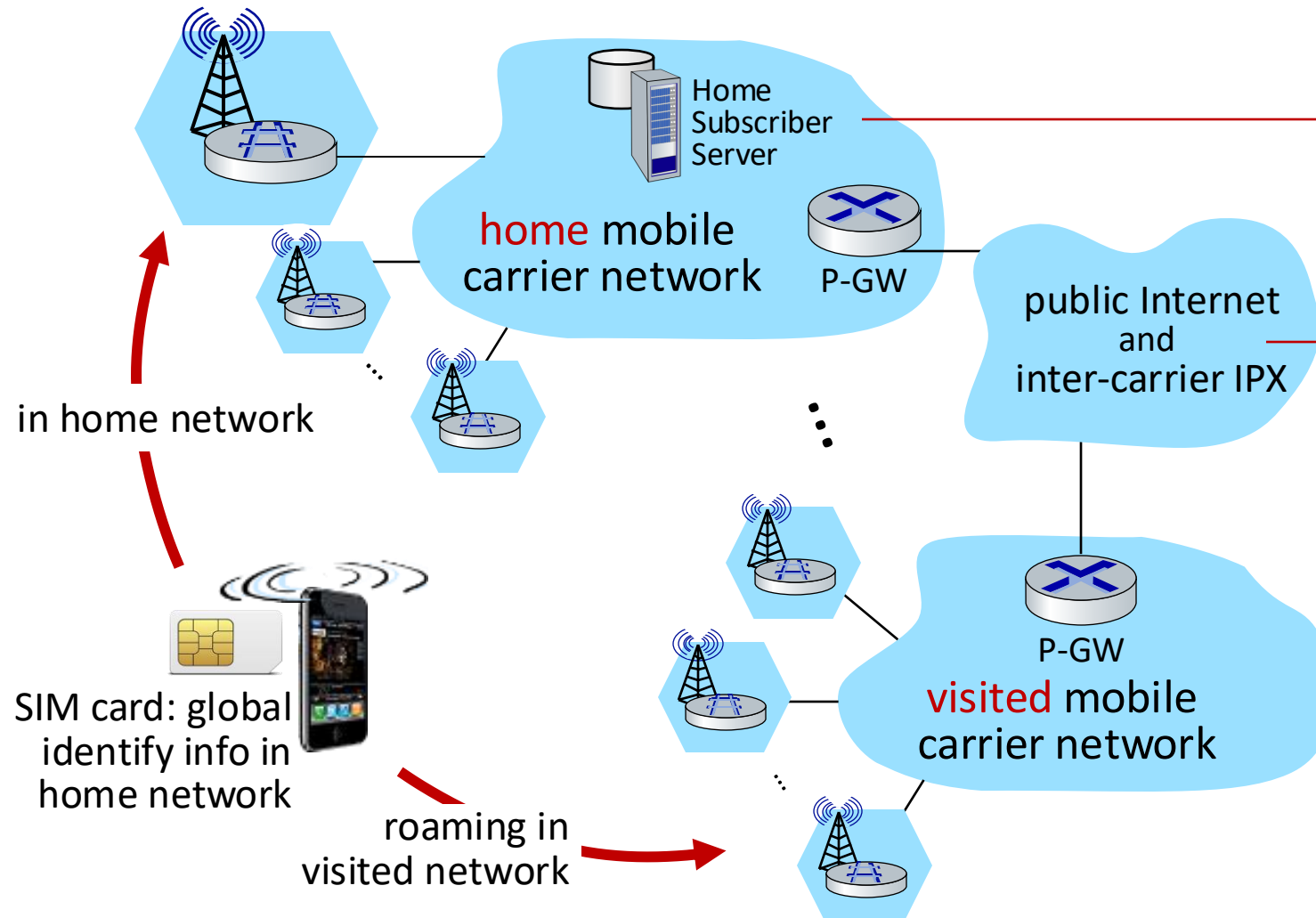
LTE mobiles: sleep modes



as in WiFi, Bluetooth: LTE mobile may put radio to “sleep” to conserve battery:

- **light sleep:** after 100's msec of inactivity
 - wake up periodically (100's msec) to check for downstream transmissions
- **deep sleep:** after 5-10 secs of inactivity
 - mobile may change cells while deep sleeping – need to re-establish association

Global cellular network: a network of IP networks



home network HSS:

- identify & services info, while in home network and roaming

all IP:

- carriers interconnect with each other, and public internet at exchange points
- legacy 2G, 3G: not all IP, handled otherwise

On to 5G!

- **goal:** 10x increase in peak bitrate, 10x decrease in latency, 100x increase in traffic capacity over 4G
- 5G NR (new radio):
 - two frequency bands: FR1 (450 MHz–6 GHz) and FR2 (24 GHz–52 GHz): millimeter wave frequencies
 - not backwards-compatible with 4G
 - MIMO: multiple directional antenna
- **millimeter wave frequencies:** much higher data rates, but over shorter distances
 - pico-cells: cells diameters: 10-100 m
 - massive, dense deployment of new base stations required